

Chapter purpose

This opening neurology chapter is a practical internal-medicine framework for translating a neurologic complaint into a lesion location, urgency tier, first diagnostic test, and initial management pathway. It emphasizes symptom-to-location, location-to-differential, differential-to-test, and test-to-action relationships for future graph-RAG extraction.

Why this matters clinically

Neurologic complaints are common, time-sensitive, and easy to mis-triage. A patient with dizziness may have benign vestibular neuritis or posterior circulation stroke. A patient with headache may have migraine or subarachnoid hemorrhage. A patient with weakness may have cortical stroke, cervical myelopathy, Guillain-Barre syndrome, myasthenia gravis, or electrolyte-driven periodic paralysis. The first task is not to name every possible diagnosis; it is to decide whether the pattern is focal or diffuse, central or peripheral, static or progressive, and dangerous or safe to evaluate more slowly.

A useful mental model is to ask four questions before ordering tests: Is there altered consciousness? Is there a focal neurologic deficit? Is there meningeal irritation or increased intracranial pressure? Is the problem episodic or fixed? These questions separate metabolic encephalopathy, stroke, seizure, meningitis, mass lesion, migraine, neuromuscular disease, and functional mimics more efficiently than a long undirected review of symptoms.

Core neurologic localization map

Localization is the neurologic equivalent of organ-system diagnosis. Internal medicine often asks whether a syndrome is cardiac, pulmonary, renal, endocrine, infectious, or hematologic. Neurology asks whether the lesion is in cortex, subcortex, brainstem, cerebellum, spinal cord, root, plexus, peripheral nerve, neuromuscular junction, or muscle. A single sign may be misleading, but combinations of signs create a lesion fingerprint.

Anatomic level	High-yield clinical pattern	Examples and diagnostic implications
Cerebral cortex	Aphasia, neglect, visual field cut, cortical sensory loss, seizure, gaze preference, contralateral face/arm/leg weakness depending on vascular territory.	Think stroke, tumor, seizure focus, abscess, encephalitis. Aphasia usually localizes to dominant hemisphere; neglect usually localizes to nondominant parietal cortex.
Subcortical/internal capsule	Pure motor or pure sensory deficits without cortical signs; dense contralateral weakness may occur with small lesions.	Think lacunar infarct from small vessel disease, demyelination, hypertensive hemorrhage. Small location can cause large deficit because fibers are compact.
Brainstem	Crossed findings: ipsilateral cranial nerve deficit with contralateral body weakness/sensory loss; diplopia, dysarthria, dysphagia, vertigo, ataxia.	Posterior circulation stroke is the cannot-miss diagnosis. Brainstem lesions can threaten airway and consciousness because reticular activating system and cranial nerve nuclei are nearby.
Cerebellum	Ataxia, dysmetria, intention tremor, nystagmus, wide-based gait, vertigo with inability to walk.	Differentiate vestibular neuritis from cerebellar stroke. Severe truncal ataxia or direction-changing nystagmus is concerning for central pathology.
Spinal cord	Bilateral weakness or sensory loss below a level, hyperreflexia below lesion, bowel/bladder dysfunction, sensory level.	Think compression, transverse myelitis, spinal cord infarct, epidural abscess. New cord syndrome often needs urgent MRI.
Nerve root	Pain radiating in dermatomal distribution, weakness in myotomal pattern, reflex loss at root level.	Radiculopathy from disc disease, foraminal stenosis, zoster, diabetic radiculoplexus neuropathy.
Peripheral nerve	Deficit in named nerve territory; distal sensory loss, reduced reflexes, weakness in nerve distribution.	Mononeuropathy, entrapment, diabetic neuropathy, vasculitic neuropathy, toxic/metabolic neuropathy.
Neuromuscular junction	Fatigable weakness, ocular/bulbar involvement, normal sensation, reflexes often preserved.	Myasthenia gravis or botulism. Respiratory failure can occur despite normal pulse oximetry early.
Muscle	Proximal symmetric weakness, myalgias, elevated CK in many myopathies, normal sensation.	Inflammatory myopathy, endocrine myopathy, statin-associated myopathy, rhabdomyolysis, muscular dystrophy.

Five-step bedside neurologic reasoning algorithm

1. Stabilize first: airway, breathing, circulation, glucose, oxygenation, temperature, and immediate trauma or toxic exposure risks. Check point-of-care glucose in any altered or weak patient.
2. Define tempo: hyperacute seconds-to-minutes suggests vascular, seizure, migraine aura, syncope, or functional event; hours-to-days suggests infection, inflammation, demyelination, toxic-metabolic disease, or evolving stroke; weeks-to-months suggests tumor, degenerative disease, chronic compression, nutritional deficiency, or systemic illness.
3. Decide focal versus diffuse: focal deficits imply structural localization until proven otherwise; diffuse confusion without focal signs often implies metabolic, toxic, infectious, endocrine, medication, or systemic disease.
4. Localize by positive and negative findings: weakness pattern, reflexes, sensory level, cranial nerves, gait, language, visual fields, coordination, and autonomic signs.
5. Choose a first test based on danger and localization: noncontrast head CT for acute stroke/bleed concern, MRI for posterior fossa/spinal cord/subacute lesions, LP for meningitis/subarachnoid workup after mass effect is reasonably excluded, EEG for unexplained episodic altered awareness or nonconvulsive seizure concern.

Mental status and altered consciousness

Altered mental status is a syndrome, not a diagnosis. The first branch point is whether the patient is globally impaired or has a focal deficit masquerading as confusion. Delirium is fluctuating inattention with impaired arousal or cognition; dementia is chronic decline without major fluctuation; coma is unarousable unresponsiveness; aphasia may look like confusion but is a focal cortical language deficit.

Pattern	Clues	Common causes	First moves
Delirium	Acute onset, fluctuating attention, sleep-wake disruption, hallucinations may occur.	Infection, hypoxia, medications, withdrawal, renal/hepatic failure, electrolyte disorders, pain, urinary retention, constipation.	Vitals, glucose, oxygenation, medication review, CBC/CMP, UA/cultures if indicated, CXR if respiratory signs, treat cause.
Coma/stupor	Impaired arousal; may have abnormal pupils, respiration, motor posturing.	Toxic-metabolic encephalopathy, CNS infection, hemorrhage, ischemia, seizure/postictal state, hypoglycemia, hypoxia.	ABCs, glucose, naloxone when opioid possible, thiamine before glucose if malnutrition risk, CT head if focal signs/trauma/unknown cause.
Aphasia mistaken for confusion	Cannot name, repeat, comprehend, or produce fluent language; attention may be intact.	Dominant hemisphere stroke, tumor, seizure/postictal aphasia, migraine aura.	Stroke evaluation if acute; language exam separates aphasia from delirium.
Nonconvulsive seizure/status	Persistent altered awareness after a seizure, subtle eye deviation/twitching, unexplained coma, fluctuating spells.	Epilepsy, stroke, hemorrhage, infection, metabolic derangement, drug toxicity.	EEG when suspicion persists; treat emergently if status epilepticus likely.
Psychiatric/functional mimic	Inconsistent exam, preserved protective responses, normal attention between episodes.	Functional neurologic disorder, catatonia, intoxication, severe mood/psychotic disorders.	Exclude dangerous neurologic/metabolic causes before anchoring on psychiatric explanation.

Trap

Do not label an older hospitalized patient as demented based on acute confusion. Delirium is common, dangerous, and frequently reversible. Conversely, do not assume a young patient with agitation has a psychiatric condition without checking glucose, oxygenation, toxidromes, infection, seizure, and head trauma risk.

Weakness: central, peripheral, neuromuscular junction, or muscle?

Weakness should be described by distribution, symmetry, reflexes, tone, sensory involvement, cranial/bulbar involvement, fatigability, and tempo. -Generalized weakness- is often fatigue, dyspnea, anemia, endocrine disease, depression, deconditioning, or medication effect. True neurologic weakness means impaired power on confrontational testing or functional tasks such as rising from a chair, dorsiflexing the ankle, or abducting the shoulder.

Feature	Upper motor neuron / central	Lower motor neuron / peripheral	NMJ	Muscle
Tone/reflexes	Increased tone, hyperreflexia, clonus, Babinski after acute phase.	Decreased tone, hyporeflexia/areflexia, fasciculations possible.	Reflexes usually normal until severe; sensation normal.	Reflexes preserved until severe weakness; sensation normal.
Distribution	Hemiparesis, paraparesis, quadriparesis; often with sensory level or cortical signs.	Root/nerve/length-dependent pattern; distal sensory symptoms common.	Ocular/bulbar/proximal fatigable weakness.	Proximal symmetric weakness; neck flexor weakness may occur.
Sensory findings	May have cortical sensory loss or sensory level.	Common: numbness, paresthesia, neuropathic pain.	Absent sensory loss.	Absent sensory loss.
Tempo clues	Sudden stroke/hemorrhage; subacute cord compression or demyelination.	Hours-days GBS; chronic diabetic/toxic neuropathy; acute compression.	Fluctuates with use; worsens later in day in myasthenia.	Weeks-months inflammatory/endocrine; acute rhabdomyolysis/toxic.
Key tests	CT/MRI brain; MRI spine when cord signs.	EMG/NCS, B12, A1c, SPEP, TSH, HIV/HCV when indicated.	AChR/MuSK antibodies, bedside respiratory mechanics, EMG.	CK, aldolase, TSH, ESR/CRP, myositis antibodies, EMG/MRI/biopsy if needed.

Sensory loss and pain localization

Sensory symptoms are useful when they obey anatomy. A sensory level suggests spinal cord disease; a dermatomal band suggests root disease; stocking-glove loss suggests length-dependent polyneuropathy; a named nerve territory suggests mononeuropathy; patchy painful multifocal deficits suggest mononeuritis multiplex; hemisensory loss with face involvement suggests brain or thalamus.

Pattern	Localization	High-yield associations
Stocking-glove numbness/burning	Length-dependent polyneuropathy	Diabetes, alcohol, B12 deficiency, chemotherapy, HIV, paraproteinemia, hypothyroidism.
Dermatomal pain with reflex change	Nerve root	Radiculopathy from disc disease/stenosis; zoster if painful vesicular rash.
Sensory level on trunk	Spinal cord	Compression, transverse myelitis, epidural abscess, B12/copper deficiency, spinal cord infarct.
Ipsilateral face and contralateral body sensory loss	Brainstem	Lateral medullary/pontine syndromes, demyelination, tumor.
Loss of vibration/proprioception with ataxia	Dorsal columns or large fibers	B12 deficiency, tabes dorsalis, sensory neuronopathy, large-fiber neuropathy.
Painful asymmetric multifocal deficits	Multiple peripheral nerves	Vasculitis, diabetes, sarcoid, amyloid, infections, malignancy.

Cranial nerve and brainstem screen

Cranial nerve findings are disproportionately valuable because they narrow localization quickly. Diplopia, dysarthria, dysphagia, facial weakness, Horner syndrome, abnormal pupils, and vertigo should be interpreted with long-tract signs and cerebellar findings. A single isolated cranial mononeuropathy can be microvascular, compressive, inflammatory, infectious, or aneurysmal depending on the nerve and pupil involvement.

Finding	Localization clue	Do not miss
Ptosis + down-and-out eye; pupil involved	CN III parasympathetic fibers often compressed externally	Posterior communicating artery aneurysm until proven otherwise.
Horizontal diplopia worse looking to affected side	CN VI palsy or increased intracranial pressure	Mass/ICP if bilateral or with papilledema; microvascular if isolated in diabetic patient.
Facial droop sparing forehead	Contralateral corticobulbar tract	Stroke pattern.
Complete unilateral facial weakness including forehead	Ipsilateral peripheral CN VII	Bell palsy, Lyme, Ramsay Hunt, tumor, stroke mimic if other deficits.
Dysphagia, hoarseness, uvula deviation	CN IX/X or medulla	Brainstem stroke, mass, neuromuscular disease, aspiration risk.
Vertigo with direction-changing nystagmus or severe truncal ataxia	Central vestibular/cerebellar pathway	Posterior circulation stroke.

Headache framework: primary, secondary, and emergency patterns

Most headaches are primary headache disorders, but internal medicine must first recognize secondary headache red flags. Primary headaches include migraine, tension-type headache, cluster headache, and medication-overuse headache. Secondary headaches are caused by another process such as hemorrhage, meningitis, mass lesion, venous sinus thrombosis, temporal arteritis, hypertensive emergency, cervical artery dissection, glaucoma, or medication/toxin exposure.

Red flag	Why it matters	Typical first action
Thunderclap or maximal intensity within 1 hour	Subarachnoid hemorrhage, reversible cerebral vasoconstriction, dissection, venous thrombosis.	Noncontrast head CT urgently; LP and/or vascular imaging if suspicion persists.
New focal neurologic deficit, seizure, papilledema, altered mental status	Mass, stroke, hemorrhage, infection, elevated ICP.	Urgent neuroimaging; avoid LP until mass effect/obstructive hydrocephalus risk addressed.
Fever, neck stiffness, immunosuppression	Meningitis, encephalitis, abscess.	Blood cultures and empiric antimicrobials promptly; CT before LP if focal deficit, papilledema, seizure, immunocompromise, or depressed consciousness.
Age >50 with new headache, jaw claudication, visual symptoms	Giant cell arteritis.	ESR/CRP, high-dose glucocorticoids if suspected, temporal artery ultrasound/biopsy depending resources.
Pregnancy/postpartum or hypercoagulability	Cerebral venous sinus thrombosis, preeclampsia/eclampsia, pituitary apoplexy.	MRI/MRV or CT/CTV depending urgency; check BP/urine protein.
Cancer, HIV, transplant, anticoagulation, trauma	Mass, opportunistic infection, hemorrhage.	Lower threshold for imaging and broader infectious/malignancy workup.

Primary headache clue

Migraine is often unilateral and throbbing with nausea, photophobia, phonophobia, and disability; aura may produce transient visual, sensory, or language symptoms. Tension-type headache is bilateral, pressing, and less disabling. Cluster headache is severe unilateral orbital/temporal pain with autonomic features such as tearing, rhinorrhea, ptosis, or miosis.

Seizure, syncope, and transient neurologic events

Transient loss of consciousness or transient neurologic symptoms require separation of seizure, syncope, TIA, migraine aura, metabolic events, and functional episodes. The key is the before-during-after story: prodrome, posture, triggers, movements, duration, recovery, tongue injury, incontinence, focal deficits, and medications.

Feature	Seizure more likely	Syncope more likely	TIA/migraine more likely
Prodrome	Aura: rising epigastric sensation, déjà vu, focal sensory/visual symptoms.	Lightheadedness, nausea, warmth, tunnel vision, diaphoresis.	TIA abrupt negative deficit; migraine gradual spreading positive symptoms.
Trigger/posture	Often unprovoked; sleep deprivation, alcohol withdrawal, flashing lights in susceptible patients.	Standing, dehydration, pain, micturition, cough, exertion, arrhythmia.	Vascular risk factors for TIA; migraine history or triggers.
Movements	Rhythmic tonic-clonic activity, forced head/eye deviation, focal automatisms.	Brief myoclonic jerks can occur after cerebral hypoperfusion.	Usually no loss of consciousness; aura can mimic focal seizure.
Recovery	Postictal confusion, headache, myalgia, focal Todd paralysis possible.	Rapid recovery when supine, minimal confusion.	TIA resolves without postictal state; migraine aura often followed by headache.
High-yield tests	Glucose/electrolytes, antiseizure drug levels if relevant, EEG, MRI brain when indicated.	ECG, orthostatics, telemetry, echo/stress testing if cardiac clues.	Brain/vascular imaging, ECG/telemetry, risk-factor workup; MRI DWI preferred for TIA when timely.

Epilepsy is usually diagnosed after two unprovoked seizures more than 24 hours apart, or after one unprovoked seizure with sufficiently high recurrence risk from an epileptogenic lesion, EEG abnormality, or defined epilepsy syndrome. A provoked seizure from hypoglycemia, hyponatremia, alcohol withdrawal, infection, medication toxicity, or acute stroke is treated by correcting the trigger and assessing recurrence risk.

Syncope risk triage

Syncope is transient loss of consciousness from global cerebral hypoperfusion with rapid onset, brief duration, and spontaneous complete recovery. The dangerous branch is cardiac syncope: exertional syncope, syncope while supine, palpitations before collapse, family history of sudden death, abnormal ECG, structural heart disease, severe anemia/bleeding, or persistent hypotension. Neuroimaging is usually low yield in uncomplicated syncope without focal neurologic findings; ECG and hemodynamic assessment are more important.

Lumbar puncture basics

Lumbar puncture is a diagnostic procedure for meningitis, encephalitis, subarachnoid hemorrhage after negative/indeterminate CT when suspicion persists, inflammatory/demyelinating disease, carcinomatous meningitis, and selected cases of intracranial pressure disorders. The major safety question is whether brain imaging should precede LP because of mass effect risk.

CSF pattern	Typical profile	Common associations
Bacterial meningitis	High opening pressure; neutrophilic pleocytosis; low glucose; high protein; Gram stain/culture may be positive.	<i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>Listeria</i> in older/immunocompromised/pregnant, gram-negative bacilli after neurosurgery.
Viral meningitis/encephalitis	Lymphocytic pleocytosis, normal glucose, mildly elevated protein; RBCs may suggest HSV encephalitis.	Enterovirus, HSV, VZV, West Nile, HIV seroconversion.
Tuberculous/fungal meningitis	Lymphocytic pleocytosis, low glucose, high protein; opening pressure often high.	TB, cryptococcus, coccidioides, histoplasma; risk rises with immunosuppression.
Subarachnoid hemorrhage	RBCs that do not clear or xanthochromia depending timing and assay.	Aneurysm rupture; evaluate with vascular imaging after confirmation.
Guillain-Barre syndrome	Albuminocytologic dissociation: high protein with few WBCs, often after first week.	Ascending weakness, areflexia, autonomic instability, respiratory risk.
Multiple sclerosis/inflammatory	Oligoclonal bands, elevated IgG index, mild lymphocytic pleocytosis.	Interpret with MRI and clinical dissemination in time/space.

LP before or after CT?

Consider CT before LP when there is focal neurologic deficit, papilledema, new seizure, moderate-to-severe impaired consciousness, known CNS lesion, or major immunocompromise. If bacterial meningitis is strongly suspected, do not delay antibiotics while waiting for imaging or LP logistics.

Neuroimaging selection for internal medicine

The best imaging test depends on the question. Noncontrast CT is fast and excellent for acute hemorrhage, mass effect, hydrocephalus, and emergent stroke triage. MRI is more sensitive for early ischemia, posterior fossa lesions, demyelination, encephalitis, tumor characterization, spinal cord disease, and many subacute processes. CTA/MRA evaluate arteries; CTV/MRV evaluate venous sinuses. Imaging should answer a clinical question rather than substitute for localization.

Clinical question	Preferred initial test in many settings	Reasoning
Acute focal deficit within stroke window	Noncontrast head CT plus vascular imaging as indicated	Rapidly excludes hemorrhage and guides thrombolysis/thrombectomy triage. MRI DWI is sensitive but should not delay emergent therapy.
TIA after symptoms resolved	MRI brain with DWI when available without delay; vascular imaging	DWI detects infarct and refines risk. Evaluate carotid/intracranial arteries and cardiac rhythm sources.
Thunderclap headache	Noncontrast head CT; CTA/LP depending timing and suspicion	Early CT is sensitive for SAH; negative CT does not always end workup if suspicion remains.
Posterior circulation symptoms	MRI brain DWI; CT/CTA if emergent	CT may miss early posterior fossa infarcts; do not dismiss severe vertigo/ataxia.
Suspected spinal cord compression	MRI spine with contrast when infection/malignancy possible	Cord syndromes are time-sensitive; plain films do not exclude compression.
New seizure with focal deficit or cancer/immunosuppression	MRI brain with/without contrast after acute stabilization; CT if emergent	MRI better detects structural epileptogenic lesions, tumor, infection, cortical dysplasia.
Altered mental status with focal findings or trauma	Noncontrast CT head initially; MRI if CT unrevealing and suspicion persists	Fast safety screen for bleed/mass; MRI for encephalitis, small infarcts, toxic/metabolic patterns.

Emergency neurologic triage: time-critical syndromes

Syndrome	Clues	Immediate priorities
Acute ischemic stroke	Sudden focal deficit: face/arm/leg weakness, aphasia, neglect, visual field cut, ataxia, dysarthria.	Last known well, glucose, noncontrast CT, CTA when large-vessel occlusion possible, BP/anticoagulant review, stroke team.
Intracranial hemorrhage	Sudden headache, vomiting, severe hypertension, decreased consciousness, focal deficits, anticoagulant use.	CT head, reverse anticoagulation when appropriate, BP control, neurosurgical/neurocritical care consult.
Subarachnoid hemorrhage	Thunderclap worst headache, neck stiffness, syncope, vomiting, photophobia.	CT head, CTA/LP depending context, manage BP/pain/nausea, nimodipine after aneurysmal SAH diagnosis.
Meningitis/encephalitis	Fever, headache, neck stiffness, altered mental status, seizure, rash, immunosuppression.	Blood cultures, empiric antibiotics +/- acyclovir, LP when safe; do not wait for LP if unstable.
Status epilepticus	Continuous seizure or recurrent seizures without return to baseline.	Airway, glucose/thiamine as indicated, benzodiazepine, loading antiseizure medication, EEG if nonconvulsive concern.
Spinal cord compression/epidural abscess	Back pain, fever or cancer risk, weakness, sensory level, urinary retention, saddle anesthesia.	Urgent MRI, steroids for malignant compression when appropriate, antibiotics and surgery for epidural abscess/compression.
Myasthenic/GBS respiratory failure	Bulbar weakness, fatigable ptosis/diplopia, ascending weakness/areflexia, weak cough.	Forced vital capacity/negative inspiratory force, ICU if declining, avoid respiratory depressants, disease-specific therapy.

High-yield lesion patterns and illness scripts

Illness script	Classic association	Diagnostic hinge
Broca aphasia	Nonfluent speech, impaired repetition, relatively preserved comprehension; dominant inferior frontal lesion.	Often MCA superior division stroke.
Wernicke aphasia	Fluent but nonsensical speech, impaired comprehension/repetition; dominant temporal-parietal lesion.	Can be mistaken for delirium unless language is tested.
Right parietal stroke	Left neglect, anosognosia, visuospatial deficits.	Patient may deny weakness or ignore one side.
Lateral medullary syndrome	Vertigo, nystagmus, dysphagia, hoarseness, ipsilateral facial pain/temp loss, contralateral body pain/temp loss, ataxia, Horner.	Posterior circulation stroke; dysphagia/aspiration risk.
Anterior spinal cord syndrome	Bilateral motor paralysis and pain/temp loss with preserved vibration/proprioception.	Spinal artery infarct, aortic disease/surgery, severe hypotension.
Brown-Sequard syndrome	Ipsilateral weakness/vibration loss with contralateral pain/temp loss below lesion.	Hemisection, trauma, tumor, MS.
Guillain-Barre syndrome	Ascending weakness, areflexia, paresthesias, autonomic instability after infection.	Respiratory mechanics and IVIG/plasma exchange; CSF protein may lag.
Myasthenia gravis	Fatigable ocular/bulbar/proximal weakness, normal sensation.	AChR/MuSK antibodies; avoid drugs that worsen MG; monitor FVC in crisis.
Temporal arteritis	Age >50, new headache, jaw claudication, visual symptoms, polymyalgia rheumatica.	Treat immediately to prevent vision loss; ESR/CRP support but do not delay steroids.

Common diagnostic traps

- Posterior circulation stroke may present as isolated dizziness, vomiting, gait inability, diplopia, dysarthria, or truncal ataxia. A normal early CT does not exclude it.
- Aphasia can look like confusion. Test naming, repetition, comprehension, fluency, reading, and writing before labeling a patient delirious.
- Syncope can have brief jerks. Convulsive movements alone do not prove seizure; postictal course and triggers matter.
- Normal oxygen saturation does not exclude impending neuromuscular respiratory failure. Check forced vital capacity and negative inspiratory force when MG or GBS is possible.

- A normal ESR does not absolutely exclude giant cell arteritis, and visual symptoms require urgent treatment.
- New severe back pain plus fever, IVU, bacteremia, cancer, anticoagulation, or neurologic deficit should trigger spinal emergency thinking.
- Do not perform LP in a patient with signs of mass effect or obstructive hydrocephalus before appropriate imaging, but do not delay empiric meningitis treatment in an unstable patient.
- Medication toxicity is neurologic: sedatives, anticholinergics, serotonergic drugs, dopamine blockers, lithium, valproate, cefepime, fluoroquinolones, metronidazole, and immunotherapies can all produce neurologic syndromes.

Practice questions

1. A 72-year-old with atrial fibrillation has sudden right arm weakness and nonfluent speech. Glucose is normal. What is the first diagnostic priority?

Answer: Treat as acute stroke. Establish last known well, activate stroke pathway, obtain noncontrast head CT to exclude hemorrhage, and obtain vascular imaging when large-vessel occlusion is possible. Aphasia localizes to dominant cortex, often left MCA territory.

2. A 29-year-old has ascending leg weakness 10 days after diarrheal illness. Reflexes are absent. Sensation is mildly abnormal. What monitoring matters immediately?

Answer: Guillain-Barre syndrome is likely. Monitor respiratory mechanics (forced vital capacity and negative inspiratory force) and autonomic instability. CSF may show high protein with few cells, but respiratory triage should not wait for LP.

3. A 62-year-old has new thunderclap headache and vomiting. Neurologic exam is nonfocal. What should be considered even if the first CT is negative?

Answer: Subarachnoid hemorrhage remains a concern depending timing and CT quality. Further evaluation may include LP for xanthochromia/RBC pattern and/or CTA to assess aneurysm or other vascular causes.

4. A hospitalized older adult becomes inattentive at night after receiving diphenhydramine and opioids. There is no focal deficit. What syndrome is most likely?

Answer: Delirium. Search for triggers: medications, infection, hypoxia, urinary retention, constipation, pain, metabolic disturbances, withdrawal, and sleep disruption.

5. A patient has fatigable ptosis, diplopia, dysphagia, and proximal weakness with normal sensation. What category of disease is suggested?

Answer: Neuromuscular junction disease, especially myasthenia gravis. Check AChR/MuSK antibodies and respiratory mechanics if bulbar or respiratory symptoms are present.

6. A patient has back pain, fever, urinary retention, and leg weakness. What is the cannot-miss diagnosis and test?

Answer: Spinal epidural abscess or compressive spinal cord lesion. Urgent MRI spine is the key test, with blood cultures and empiric antibiotics if infection is suspected.

Graph-RAG extraction map

Chapter 9A: Graph-RAG extraction map and references

Graph-RAG extraction map

Altered mental status: neurologic symptom/emergency syndrome; caused by delirium, seizure, stroke, meningitis, toxic-metabolic disease; evaluated by glucose, vitals, medication review, CT/LP/EEG.

Aphasia: cortical language deficit; localizes to dominant hemisphere; helps distinguish focal cortical lesion from generalized confusion.

Upper motor neuron weakness: hyperreflexia, Babinski sign, spasticity; suggests brain or spinal cord.

Lower motor neuron weakness: hyporeflexia, atrophy, fasciculations; suggests root, plexus, nerve, or anterior horn cell process.

Thunderclap headache: vascular emergency; concern for SAH, RCVS, dissection, venous sinus thrombosis; triggers urgent CT/CTA/LP logic.

Seizure: transient neurologic event; differentiated from syncope by aura, postictal state, tongue injury, witnessed semiology; evaluated with glucose, electrolytes, EEG, MRI when indicated.

Syncope: transient loss of consciousness from cerebral hypoperfusion; cardiac red flags drive ECG, telemetry, and admission decisions.

Lumbar puncture: diagnostic procedure for meningitis, SAH workup, and inflammatory disease; CT first if mass effect risk is present.

Selected references used for clinical cross-checking

- AHA/ASA acute ischemic stroke and TIA diagnostic guidance.
- ACR Appropriateness Criteria: Headache; Altered Mental Status, Coma, Delirium, and Psychosis.
- International League Against Epilepsy seizure and epilepsy definitions/classification resources.
- American Academy of Neurology resources for first seizure and headache quality measures.
- Standard emergency neurology practice: glucose-first altered mental status, CT-before-LP risk assessment, empiric meningitis treatment urgency, and spinal cord compression triage.